Course Project Report

Title of the Project

Sl. No.	Name of Student	Roll Number	Department
1	Student Name 1	Roll No. 1	Department 1
2	Student Name 2	Roll No. 2	Department 2
3	Student Name 3	Roll No. 3	Department 3
4	Student Name 4	Roll No. 4	Department 4

1 Problem Formulation

- Present the real-world problem in precise and formal terms.
- Discuss why the problem is important, interesting, and worthy of study.
- If the project is interdisciplinary, explain the attractiveness and significance of combining ideas from multiple domains.
- Clearly define the inputs, outputs, and the expected decision or solution process.
- Characterize the problem appropriately, for example as deterministic or stochastic, static or dynamic, discrete or continuous, and constrained or unconstrained.

2 Method and Justification

Why This Method is Appropriate

- Explain why the chosen method matches the problem structure.
- Link problem properties to algorithm strengths.
- Explain why this method is suitable in terms of optimality, efficiency, interpretability, or feasibility.

Algorithm Design and Methodology

- Explain the full pipeline from input to output.
- Describe each major stage of the algorithm.

Algorithm 1 Title of Your Algorithm

```
Require: Input data / initial state / constraints
Ensure: Desired solution / goal-reaching plan / optimized output
Initialize all required variables and data structures
Construct the initial state
while termination condition is not satisfied do
    Select the next state / variable / action according to the chosen method
    Apply transition / assignment / rule / expansion
    Evaluate feasibility or quality
    if goal condition is satisfied then
        Return the solution
    end if
end while
Return failure or best available result
```

3 Implementation Details

System Design

- Explain the software structure briefly.
- Mention important modules/functions.
- Describe key data structures.

GitHub Repository

- Provide the GitHub repository link.
- Mention that the repository includes code, documentation, and README.

Repository Link: https://github.com/your-repository-link

4 Results and Performance Analysis

Experimental Setup

- Describe datasets, test cases, or scenarios.
- Mention hardware/software setup if relevant.
- Mention evaluation metrics.

Quantitative Results

- Present result tables and graphs.
- Report metrics such as time, cost, path length, accuracy, success rate, or constraint satisfaction.